

assign 01 //

site modeling

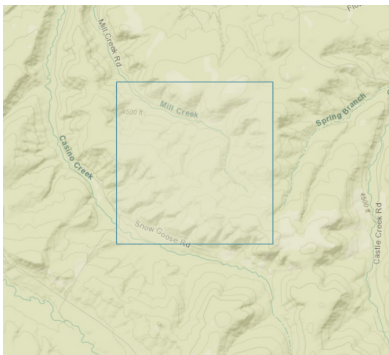
In this assignment, 2 different methods of the site modeling are explored; the first being using point clouds and the second using OSM data. In these two exercises, I learned that LiDAR data contains a immense amount of data that can be captured quickly and contains an immense amount of detail. It seems as though this type of data collection offers a mostly complete representation of site geometry without any human errors that can occur in traditional surveying.

Some of the limitations of LiDAR became apparent to me when working through the process outlined below. The primary limitation seems that the data needs to be processed and filtered to gain some understanding of the data or make the data usable in this setting. In relation to hand building models, I think that LiDAR offers the first step in understanding some of the nuances of a site that can be interpreted into architectural meaning.

workflow LiDAR topo to Revit

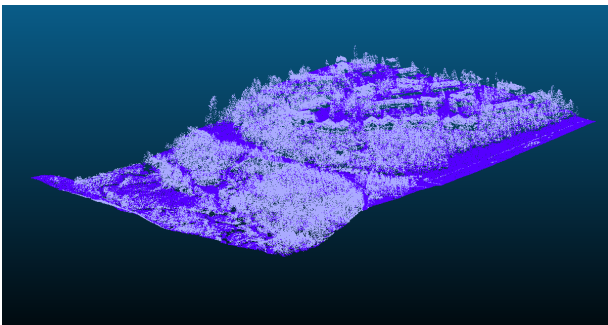
1. gather LiDAR data (point clouds)
2. upload points to cloud compare
3. construct mesh in cloud compare
4. mesh to Rhino

Using opentopography.org, the map region can be selected to download point cloud information.

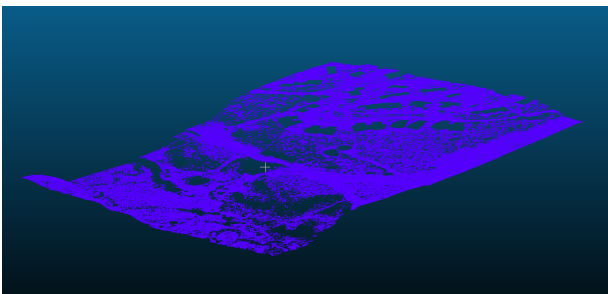


Montana Lidar Inventory is another resource for MT specific locations.

Using the .laz file from step 1, drag the file into a new instance of cloud compare.



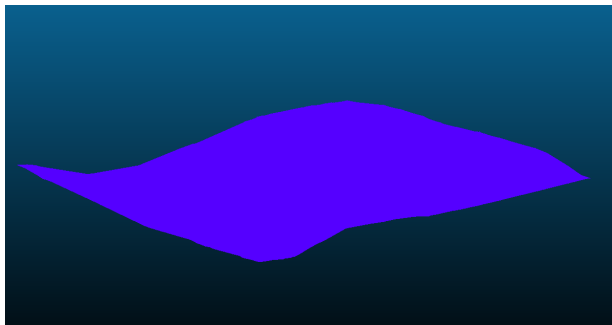
Ground points and off ground points together.



Ground points only

Once the data is in cloud compare, select the mech and use the CSF plug-in to find the ground points and off ground points

Using the cleaned up point cloud, select the ground points and use the tool drop down menu to create the mesh.

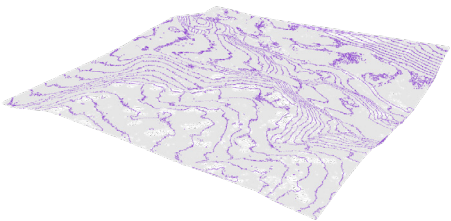


Mesh from the ground points.

Save ground points and off ground points as separate files becuae settings may vary.

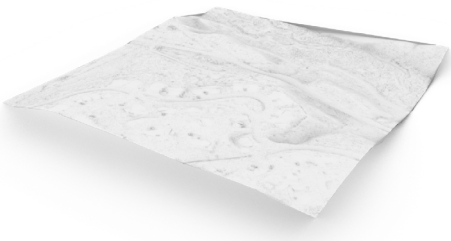
Save the mesh as an obj file that can be opened in Rhino.

Open the mesh in Rhino.



Contoured mesh

Once the mesh is in Rhino, move the mesh to the origin 0,0,0. Run the contour command and specify the distance between each of the contours.

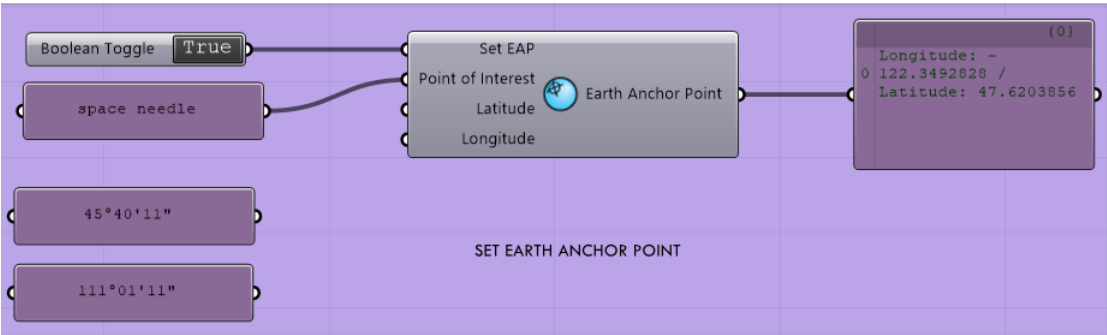


Mesh

workflow GIS modeling

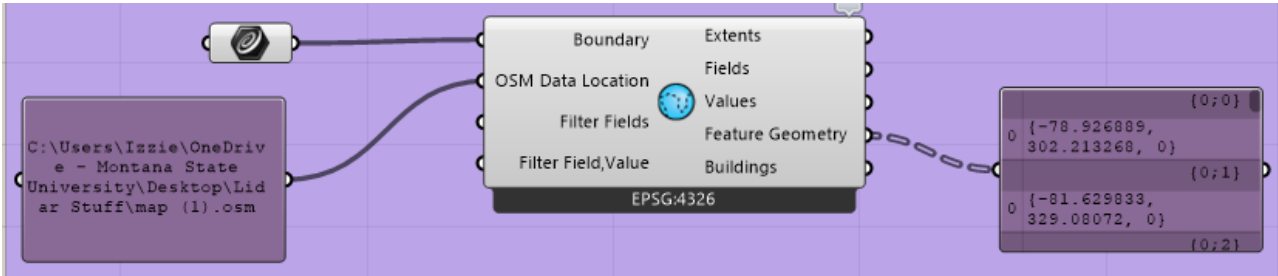
For this assignment, I worked on site building in an urban environment that was the site of our studio field trip at the start of the semester.

1. set earth anchor point



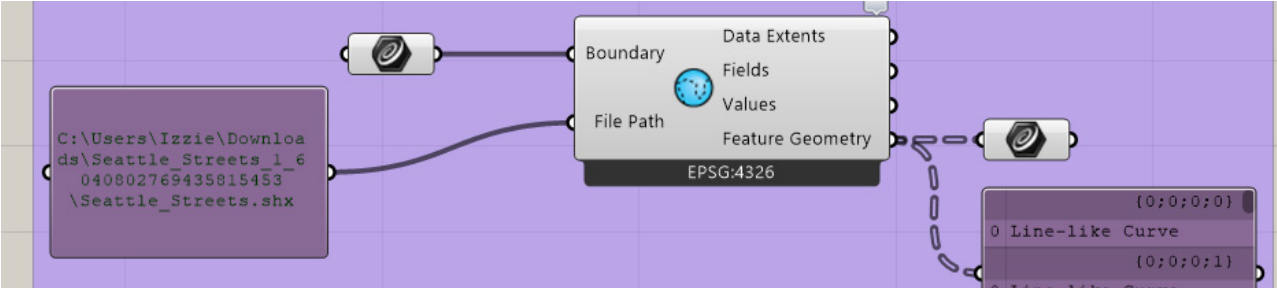
In step 1, the EAP is set using a point of interest because the space needle is at the site. In another application, the longitude and latitude can be used to set the EAP as a point of reference.

2. import OSM data



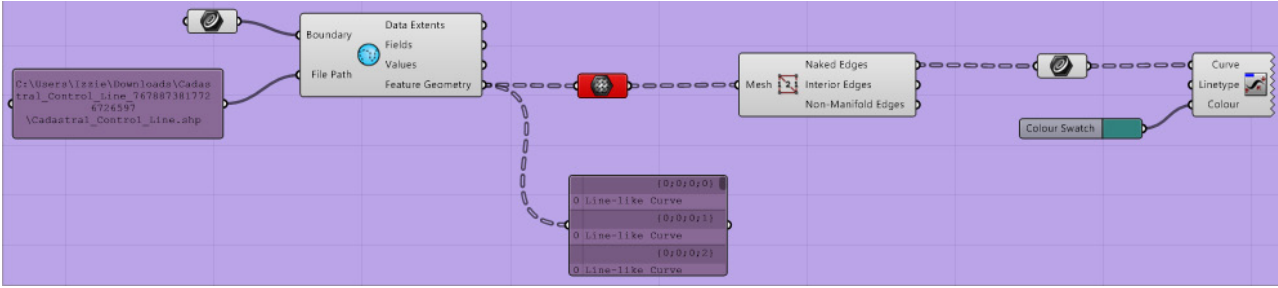
Using openstreetmaps.org, I selected the are of interest on the map and downloaded the .osm file. To import the data from the file, a curve container is set, then the file path is copied and pated into a panel. The result after this step is some building masses.

4. import data from parcels



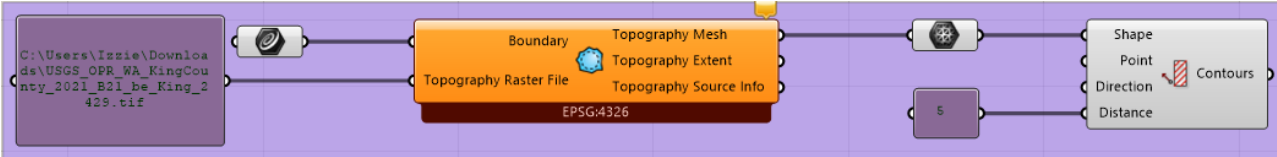
Using a shape file acquired from the city of Seattle open data portal, the roads are imported via a copy and pasted file path into a panel after a curve container has been set to the established boundary. The resulting output is paled into a curve container and a panel is added as a preview to the collected data.

3. import data from parcels



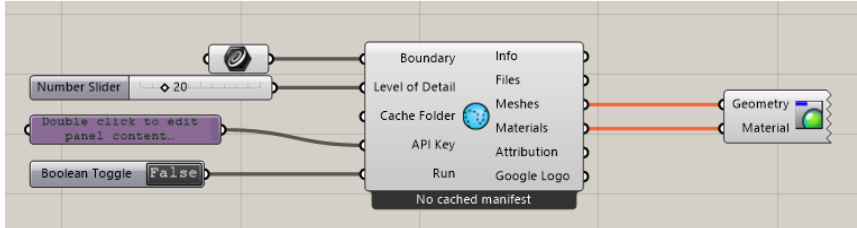
In this step, the data was downloaded from the city of Seattle GIS portal. It seems like a gis data portal is a standard offering from a city. In this step, a curve container is set to the same curve and the file path is copied and pasted into a panel similar to the last step. The output is placed to a mesh container and then into a mesh edges container to translate the naked edges into a curve container. The following components are part of a custom preview.

5. import topo



The topo data for this was acquired from the USGS national map as a .tif file type. Again a curve container sets the boundary of the model and the path is copied and pasted into a panel. The result is a mesh that is attached to a contour component and the distance is specified via a panel.

5. import topo



In this step, Google 3d tiles are added via a file path to an API key. The same boundary is set to the curve container. A # slider is added to specify the level of detail.